Project Report

Electric Vehicle Market Analysis Dashboard

Date	26 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Analysis Dashboard

1. Introduction

1.1 Project Overview

The electric vehicle (EV) industry is one of the fastest-growing sectors of the global automotive market, driven by environmental awareness, government incentives and advances in battery technology. This project uses Tableau to analyse an EV dataset covering vehicle categories, manufacturers, battery technologies, driving range, customer ratings and manufacturing locations.

The work produces nine interactive worksheets, two dashboards and a data story that give executives, business intelligence teams and researchers a clear view of annual sales, manufacturer performance, battery technology adoption, range, customer satisfaction and global production hubs.

1.2 Purpose

The purpose of the project is to turn raw EV market data into visual insights that support decisions. Specifically, it helps users to:

- Identify which vehicle categories contribute most to annual sales volume.
- Compare manufacturers against the market average.
- Evaluate battery technologies (LFP, NMC, NCA, Sodium-Ion, Solid-State) on range, rating and adoption.
- Understand where EVs are produced and how customer satisfaction varies.

2. Ideation Phase

2.1 Problem Statement

EV manufacturers, consultants and researchers hold large volumes of sales, battery, range and customer data, but it is spread across many fields and hard to interpret. Without a consolidated view, it is difficult to see which categories and battery technologies lead the market, how manufacturers compare, and what drives customer satisfaction.

Problem statement: How might we present EV sales, battery, range, production and customer-rating data in an interactive dashboard so that strategy, business intelligence and research users can quickly find insights and make data-driven decisions?

2.2 Empathy Map Canvas

The map below is based on the three user personas in the project scenario: a Market Strategy Director, a Business Intelligence Manager and a Sustainability Research Analyst.

Says	Thinks
"Which categories and manufacturers lead the market?" "Which battery technology is winning?"	Data is scattered and slow to analyse. Decisions need to be backed by evidence.
Does	**Feels**
Reviews sales and range reports, compares manufacturers, advises clients and policymakers.	Pressure to spot trends early; frustrated by static reports; confident when insights are visual and interactive.

Pains: scattered data, no single view of the market, difficulty comparing battery technologies. Gains: one interactive view, faster decisions, clear evidence for strategy and policy.

2.3 Brainstorming

Ideas considered for the dashboard, and the ones selected:

- **Sales by category**: dual-axis plot of sales volume and revenue by vehicle category (selected).
- **Manufacturer comparison**: stacked bars and a market share leaderboard with an average line (selected).
- **Battery technology view**: treemap, category vs battery heatmap and range vs rating scatter (selected).
- **Geography**: map of production hubs and export distribution (selected).
- **Customer satisfaction and impact**: pie chart of annual sales units in the CSAT and environmental impact view (selected).
- **Ideas kept for later**: price trend forecasting and charging infrastructure analysis.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User action	Need	Dashboard support
Awareness	Opens the dashboard	Quick overview of the market	Headline visuals on Dashboard 1
Exploration	Reviews categories and manufacturers	Compare performance	Category, stacked bar and leaderboard charts
Deep dive	Studies battery technology and range	Understand technology trends	Treemap, heatmap and range vs rating scatter
Insight	Looks at hubs and customer ratings	Link production and satisfaction	Map and pie chart on Dashboard 2
Decision	Shares findings	Evidence for strategy or policy	Story and Tableau Public link

3.2 Solution Requirement

Functional requirements

- Connect to the EV dataset and prepare fields (category, manufacturer, battery type, annual sales units, average range, customer rating, location).
- Build worksheets for sales, manufacturers, battery technology, range, ratings and geography.
- Combine worksheets into dashboards and a story, and publish to Tableau Public.

Non-functional requirements

- Clear, readable layout with consistent colour use.
- Interactive filtering and tooltips.
- Accurate figures consistent with the source data.
- Shareable through a public link.

3.3 Data Flow Diagram

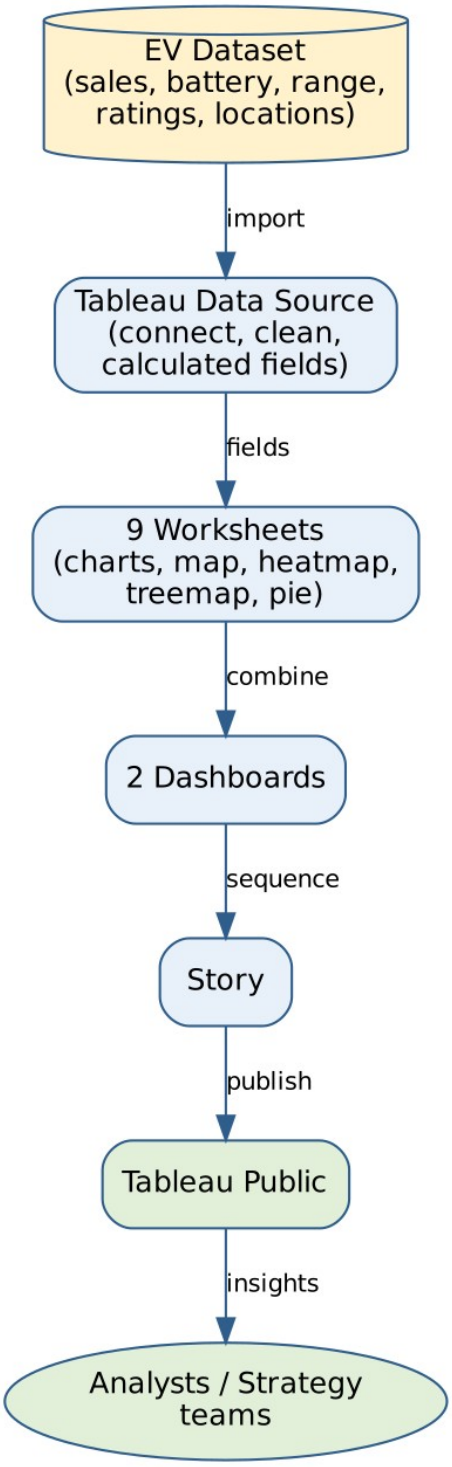


Figure 1: Data flow from dataset to published insights

3.4 Technology Stack

Component	Technology
Visualization and analytics	Tableau Desktop / Tableau Public

Data source	EV dataset (CSV / Excel)
Map layer	Tableau maps (Mapbox / OpenStreetMap)
Publishing	Tableau Public
Documentation	Microsoft Word / PDF

4. Project Design

4.1 Problem Solution Fit

The problem is the lack of a single, interactive view of the EV market. The solution matches it by consolidating sales, battery, range, rating and production data into dashboards that answer the key business questions directly, so each persona can reach its insight without manual analysis.

4.2 Proposed Solution

- Two dashboards: Dashboard 1 (sales volume, category portfolio, range vs price, manufacturer market share) and Dashboard 2 (production hubs, category vs battery type, battery degradation profile, CSAT and environmental impact).
- A data story that walks viewers through the findings in order.
- Colour-coded charts, labelled values and an average reference line for quick comparison.

4.3 Solution Architecture

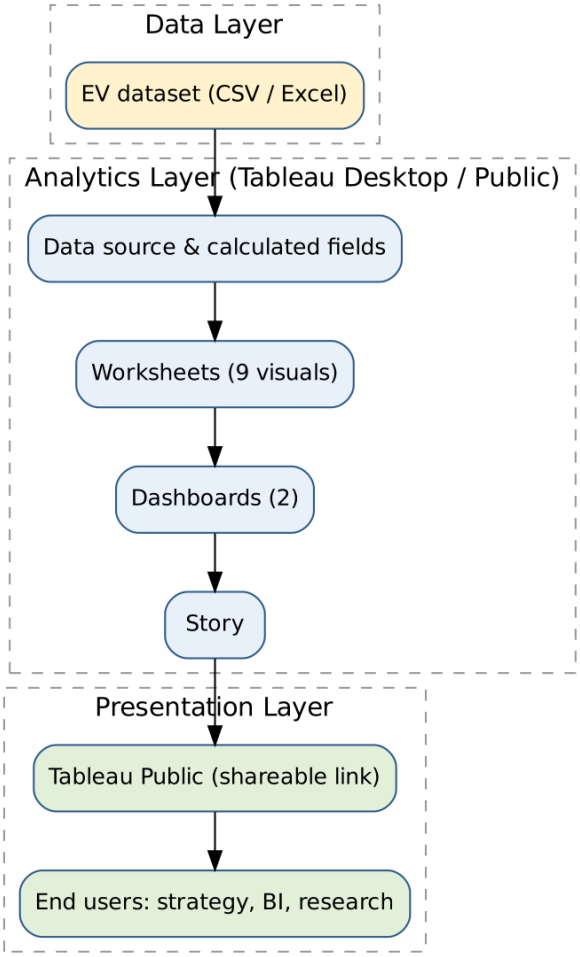


Figure 2: Solution architecture

5. Project Planning & Scheduling

5.1 Project Planning

Phase	Activities	Output
Planning	Understand scenarios and define business questions	List of business questions
Data preparation	Connect data source, check fields, create calculated fields	Clean data source
Visualization	Build 9 worksheets	Worksheets 1 to 9
Dashboard design	Arrange visuals into 2 dashboards	Dashboard 1 and Dashboard 2
Storytelling	Sequence dashboards into a story	Story
Testing and documentation	Review visuals and values, prepare reports	Project reports
Publishing	Publish to Tableau Public	Public link

6. Functional and Performance Testing

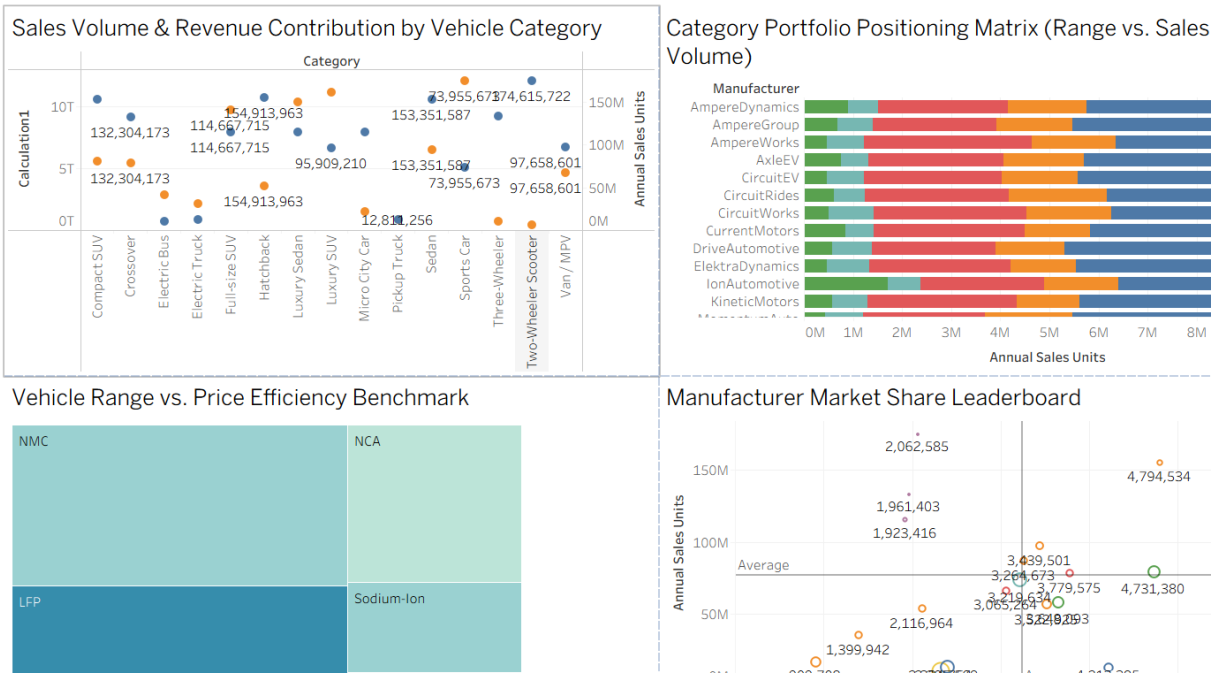
6.1 Performance Testing

S.No.	Parameter	Values
1	Data rendered	EV dataset connected as a Tableau data source
2	Data preprocessing	Fields checked and standardised before building worksheets
3	Utilization of filters	Filters applied on category, manufacturer and battery type
4	Calculation fields used	Calculation1 (sales volume and revenue contribution sheet)
5	Dashboard design	9 visualizations across 2 dashboards
6	Story design	9 visualizations across 2 story points

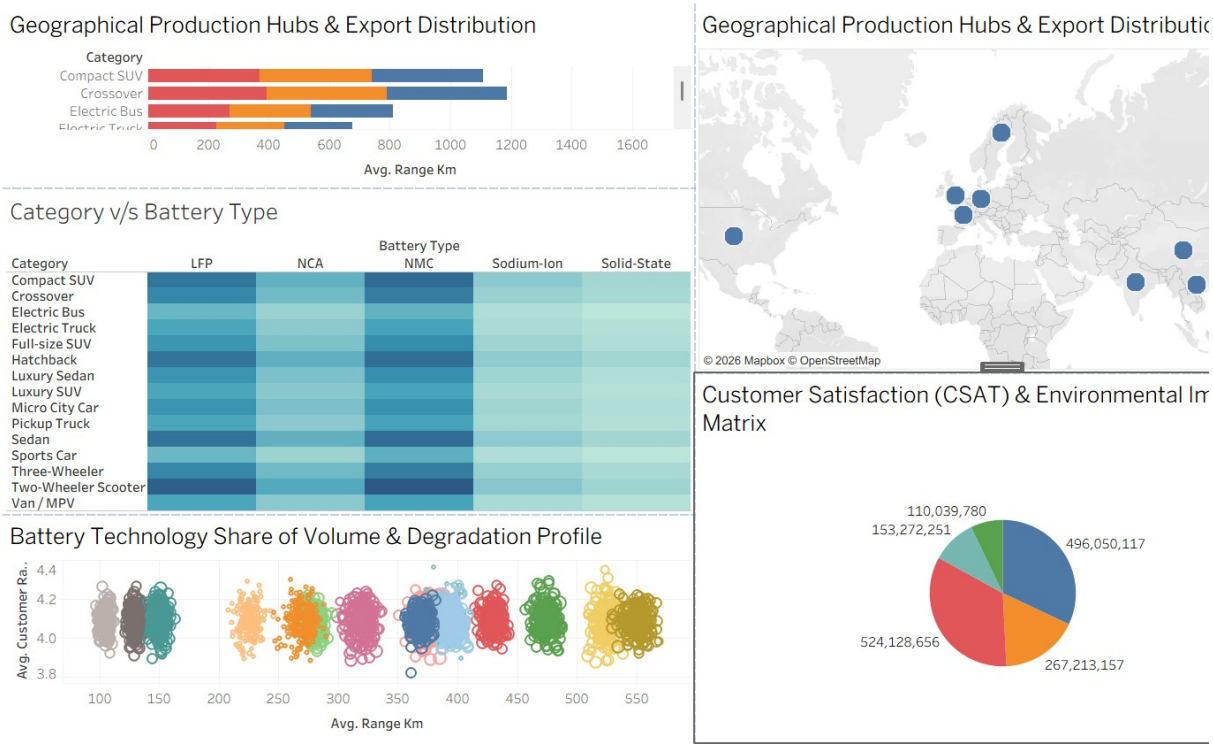
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7. Results

7.1 Output Screenshots



Dashboard 1



Dashboard 2

Key findings

- Total annual sales across the five segments is about 1.55B units; the two largest segments (about 524.1M and 496.1M) make up roughly 65.8%.
- Two-Wheeler Scooter has the highest labelled sales volume (about 174.6M units).
- NMC and NCA take the largest tiles in the range vs price treemap; LFP and NMC are the most prominent in the category vs battery heatmap.
- Crossover (about 1,180 km) and Compact SUV (about 1,100 km) lead on average range among visible categories.
- A few manufacturers stand out at about 4.79M and 4.73M units; most cluster around or below the average.
- Production hubs concentrate in Western Europe and the Nordics, with others in North America and Asia.
- Customer ratings stay between 3.8 and 4.4 even though average range varies from about 100 to 550 km.

8. Advantages & Disadvantages

Advantages	Disadvantages
Single interactive view of the EV market	Depends on the quality and coverage of the dataset
Easy comparison of categories, manufacturers and battery types	Static snapshot unless the data source is refreshed
Supports faster, evidence-based decisions	Tableau Public requires data to be public
Shareable through a public link	Dense charts can be hard to read on small screens

9. Conclusion

The project shows how Tableau can turn EV sales, battery, range, rating and production data into clear visual insights. The dashboards and story reveal which categories and battery technologies lead, how manufacturers compare and where production is concentrated, helping strategy, business intelligence and research users make informed decisions about the EV market.

10. Future Scope

- Add time-based data to analyse sales growth and forecast demand.
- Include pricing, charging infrastructure and cost-of-ownership data.
- Add more interactive filters and parameters for scenario analysis.
- Connect to a live data source for automatic refresh.
- Extend the analysis with carbon emission savings by region.

11. Appendix

Source Code (if any)

Not applicable: the project was built in Tableau without custom code (calculated fields only).

Dataset Link

[Paste dataset link here]

GitHub & Project Demo Link

[Paste GitHub link and Tableau Public dashboard / story link here]